

FIGURE 6.1

Number of plants observed
in tracts at oil spill and
control sites. The data are
displayed in stem-and-leaf
plots

Control Tracts				Oil Spill Tracts			
Mean:	38.48	000	0	Mean:	26.93		
Median:	41.50	7	0	Median:	26.00		
St. Dev:	16.37	1	1	St. Dev:	9.88		
<i>n</i> :	40	6	1	<i>n</i> :	40		
		4	2				
		9	2				
		0	3				
		55678	3				
		000111222233	4				
		57	4				
		0112344	5				
		67789	5				

TABLE 6.1

Summary statistics for oil
spill data

Descriptive Statistics						
Variable	Site Type	<i>N</i>	Mean	Median	Tr. Mean	St. Dev.
No. plants	Control	40	38.48	41.50	39.50	16.37
	Oil spill	40	26.93	26.00	26.69	9.88
Variable	Site Type	SE Mean	Minimum	Maximum	Q1	Q3
No. plants	Control	2.59	0.00	59.00	35.00	51.00
	Oil spill	1.56	5.00	52.00	22.00	33.75

summary statistics given in Table 6.1. A boxplot of the data displayed in Figure 6.2 indicates that the control sites have a somewhat greater plant density than the oil spill sites. From the summary statistics, we have that the average flora density in the control sites is $\bar{y}_{\text{Con}} = 38.48$ with a standard deviation of $s_{\text{Con}} = 16.37$. The sites within the spill region have an average density of $\bar{y}_{\text{Spill}} = 26.93$ with a standard deviation of $s_{\text{Spill}} = 9.88$. Thus, the control sites have a larger average flora density and a greater variability in flora density than the sites within the spill region. Whether these observed differences in flora density reflect similar differences in

FIGURE 6.2

Number of plants observed
in tracts at control sites (1)
and oil spill sites (2)

